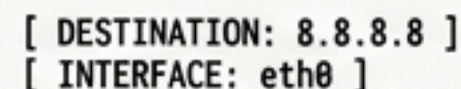


A layer-by-layer dissection of network protocols, packet captures, and points of failure.

A layer-by-layer dissection of network protocols, packet captures, and points of failure.



```
[ ENVIRONMENT: KALI LINUX ]
[ PRIMARY SENSOR: WIRESHARK ]
```

Establishing the Network Baseline

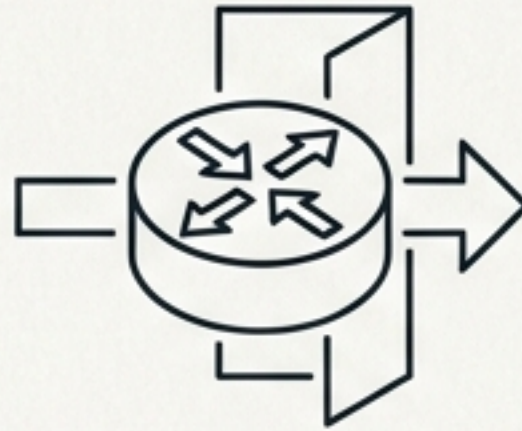
The Interface



```
$ ip a  
IP: 192.168.1.15/24
```

The Local Node & Subnet

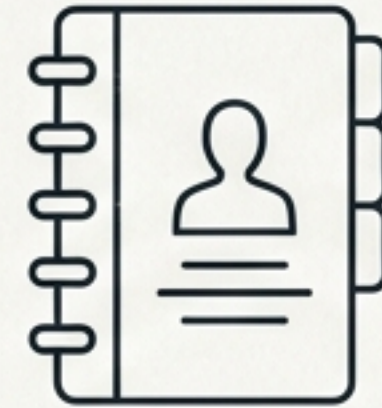
The Exit Door



```
$ ip route show  
default via 192.168.1.1
```

The Default Gateway

The Directory



```
$ cat /etc/resolv.conf  
nameserver 192.168.1.1
```

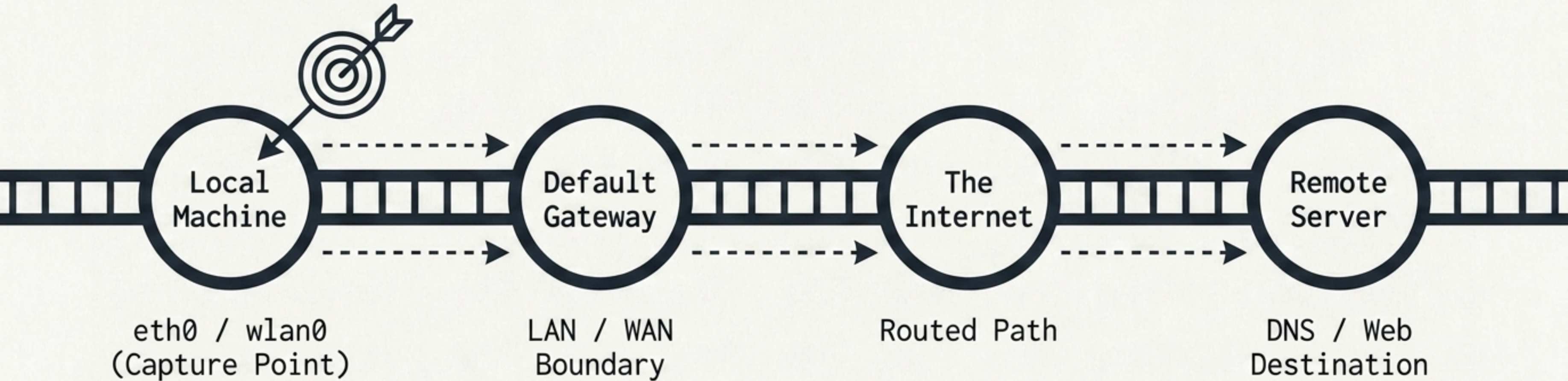
The DNS Server

Analyst Note:

On home networks, the Gateway IP and DNS Server IP are often identical. The home router acts as a forwarder for both routing and name resolution.

The Topography of a Network Tap

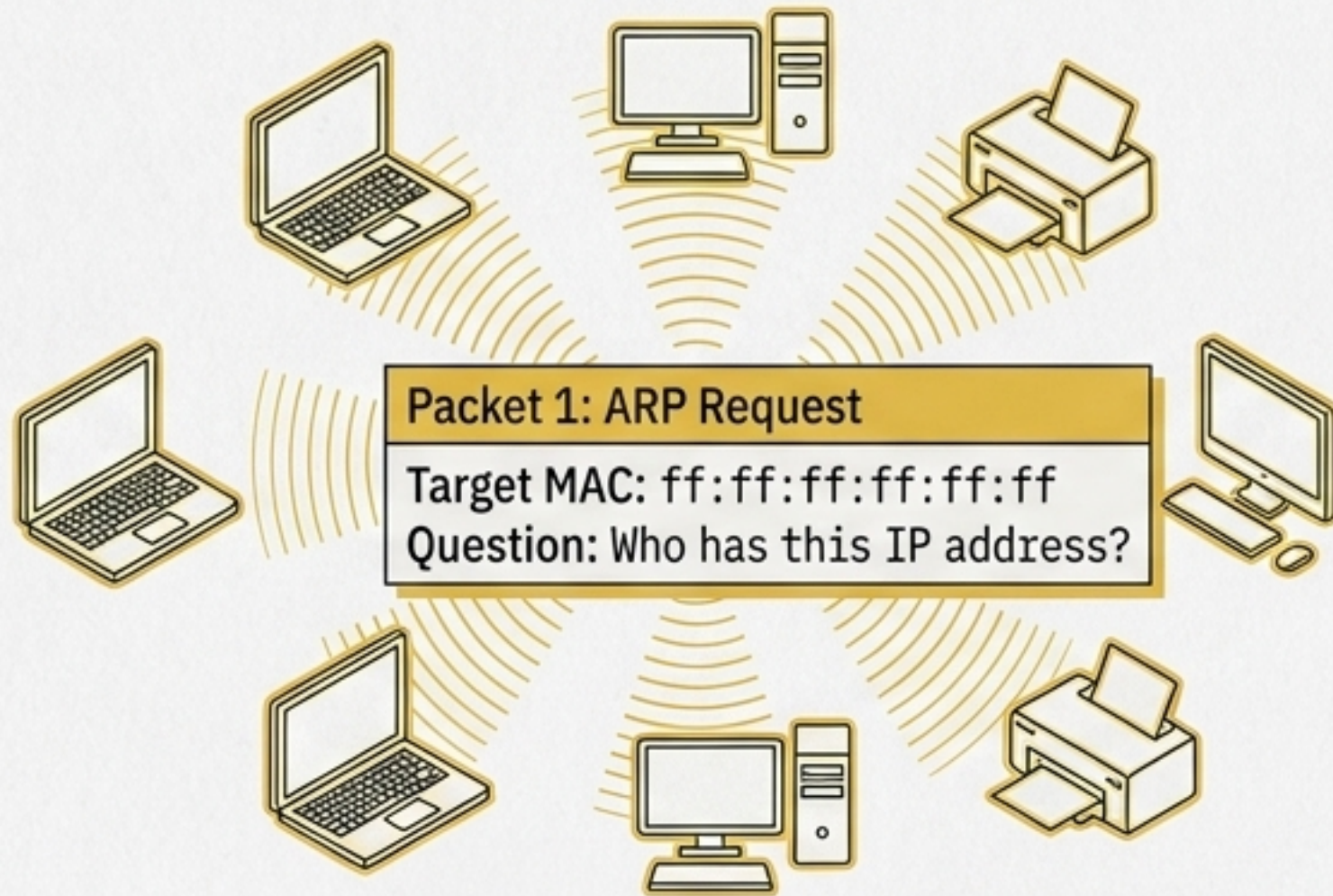
All subsequent observations are captured from the perspective of a single sensor.



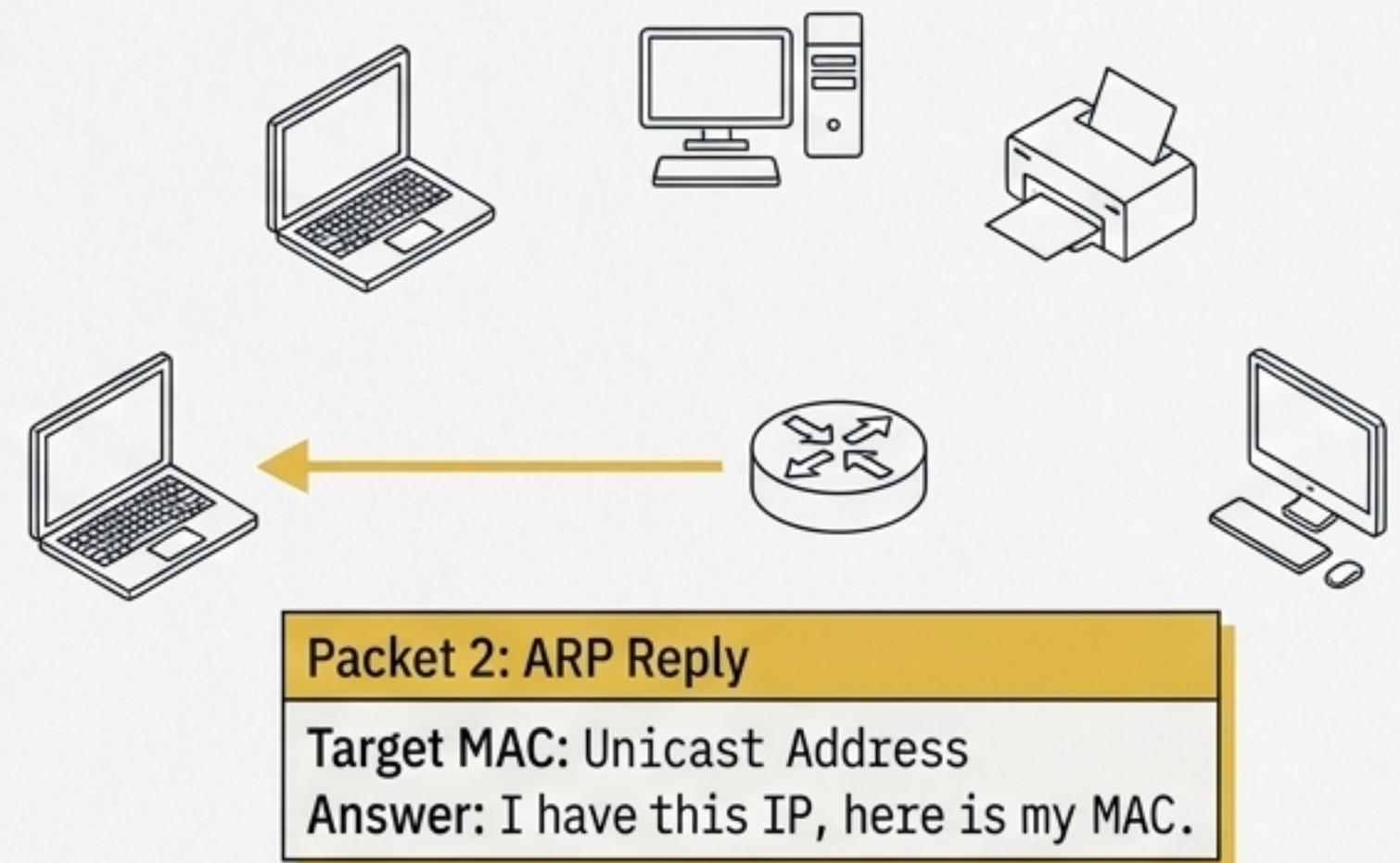
Bridging the Logical and Physical with ARP

IP addresses are logical; MAC addresses are physical. ARP bridges the gap.

Broadcast (Opcode 1)



Unicast (Opcode 2)



```
Command: ip neigh show
State: REACHABLE
Filter: arp
```

Translating Domains to Addresses

The OS uses local caching to drastically reduce network noise.



DNS Record Types

A (IPv4)	AAAA (IPv6)
<code>dig domain A</code>	<code>dig domain AAAA</code>
MX (Mail Server)	NS (Name Server)
<code>dig domain MX</code>	<code>dig domain NS</code>

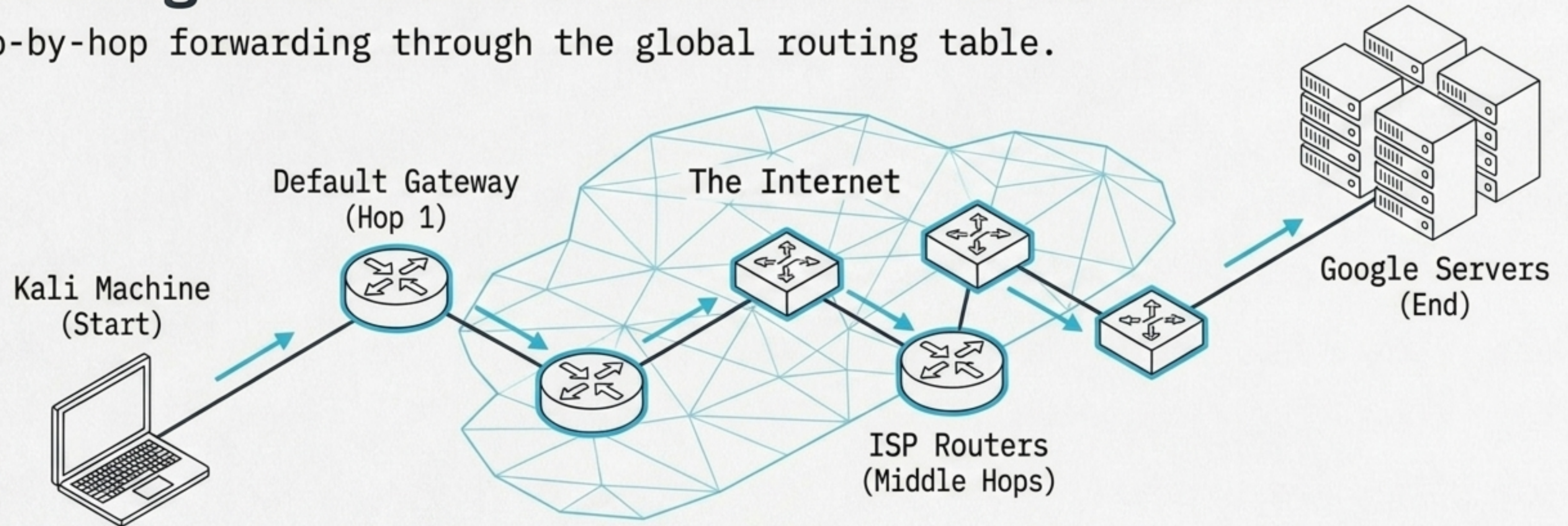
Countdown Timer



The OS will not query the network again for this exact domain until the TTL timer reaches zero.

Tracing the Path Across the Internet

Hop-by-hop forwarding through the global routing table.



```
Command: traceroute google.com
traceroute to google.com (172.217.164.110), 30 hops max, 60 byte packets
 1 192.168.1.1 (192.168.1.1)    0.526 ms  0.489 ms  0.512 ms
 2 10.0.0.1 (10.0.0.1)        5.123 ms  4.891 ms  4.956 ms
 3 203.0.113.1 (203.0.113.1)  12.456 ms * *
 4 74.125.243.1 (74.125.243.1) 18.789 ms 18.456 ms 18.912 ms
...
10 google.com (172.217.164.110) 45.123 ms 45.456 ms 45.789 ms
```



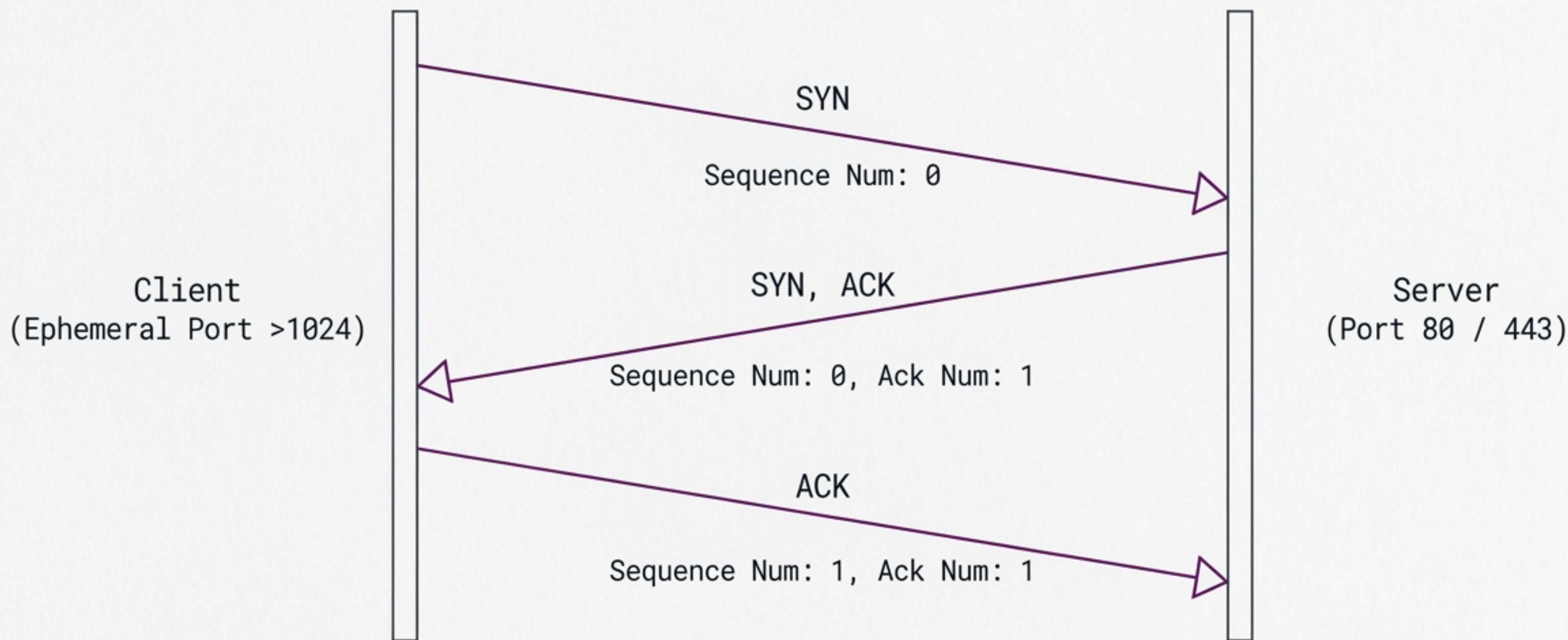
Analyst Insight

A * * timeout mid-trace does not always indicate a broken path. It often simply means an intermediary router's firewall policy is configured to drop diagnostic ICMP probes.

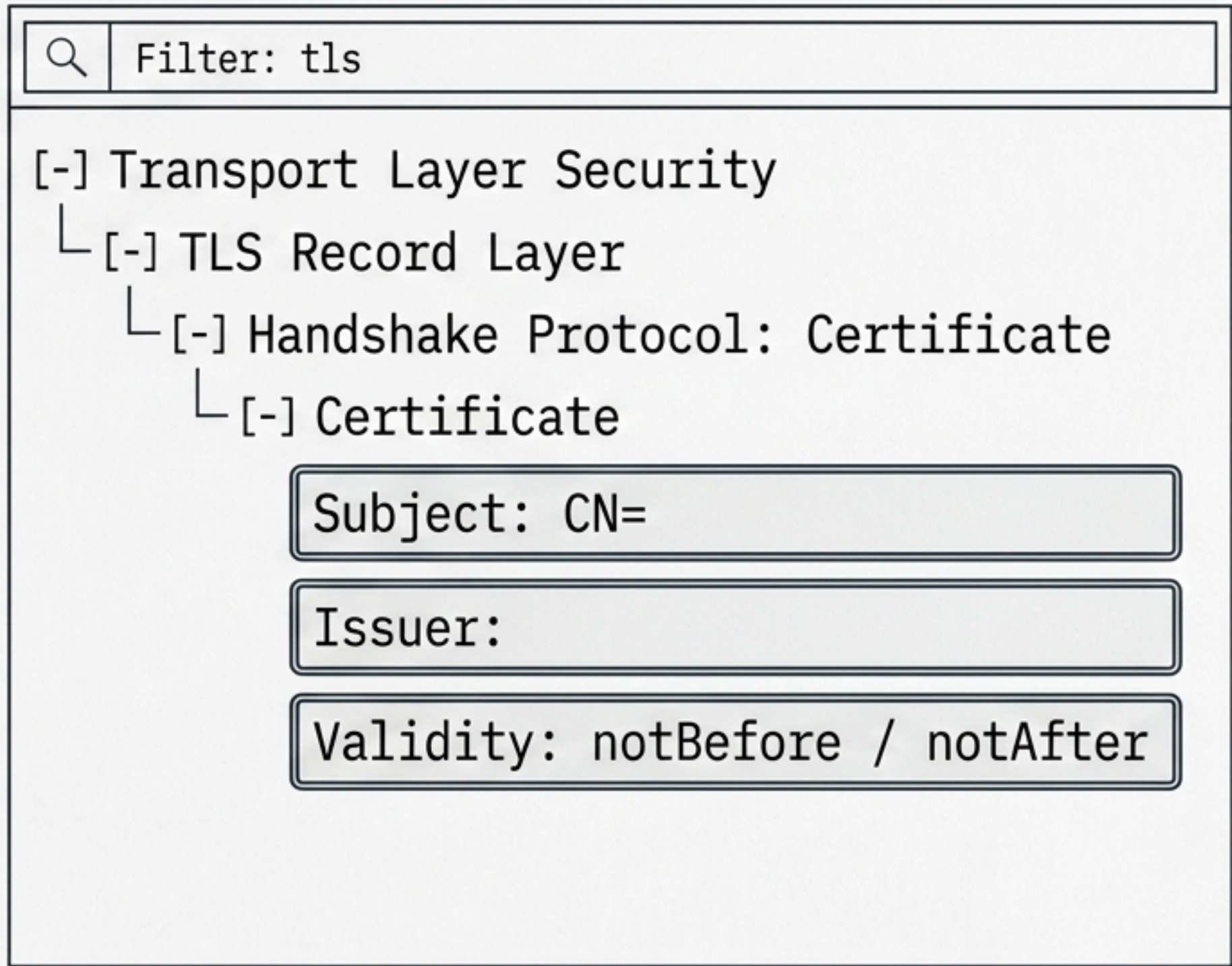
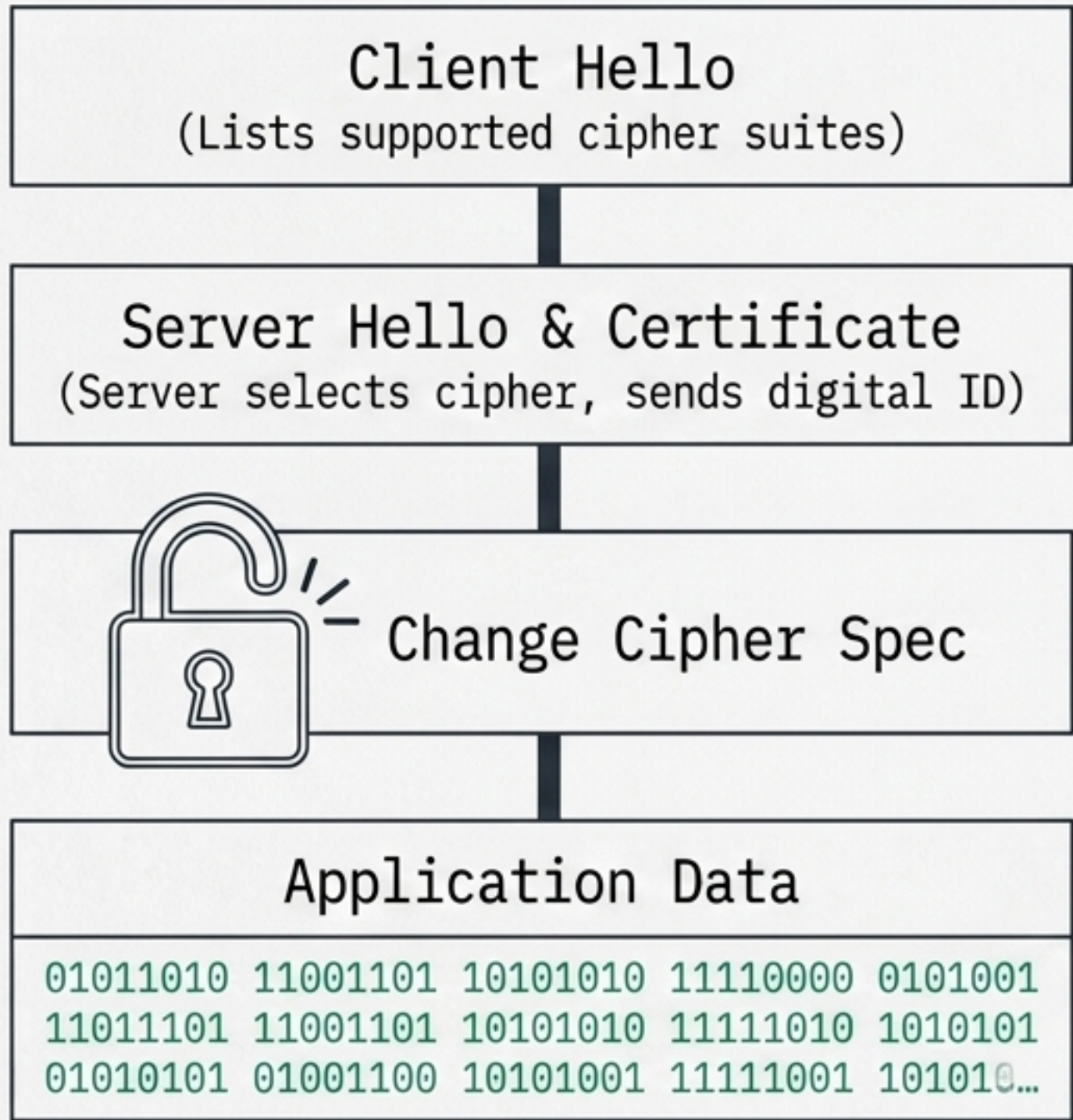
Forging a Reliable Connection

Filter: `tcp.flags.syn == 1`

The TCP 3-Way Handshake establishes temporal sequencing and trust.

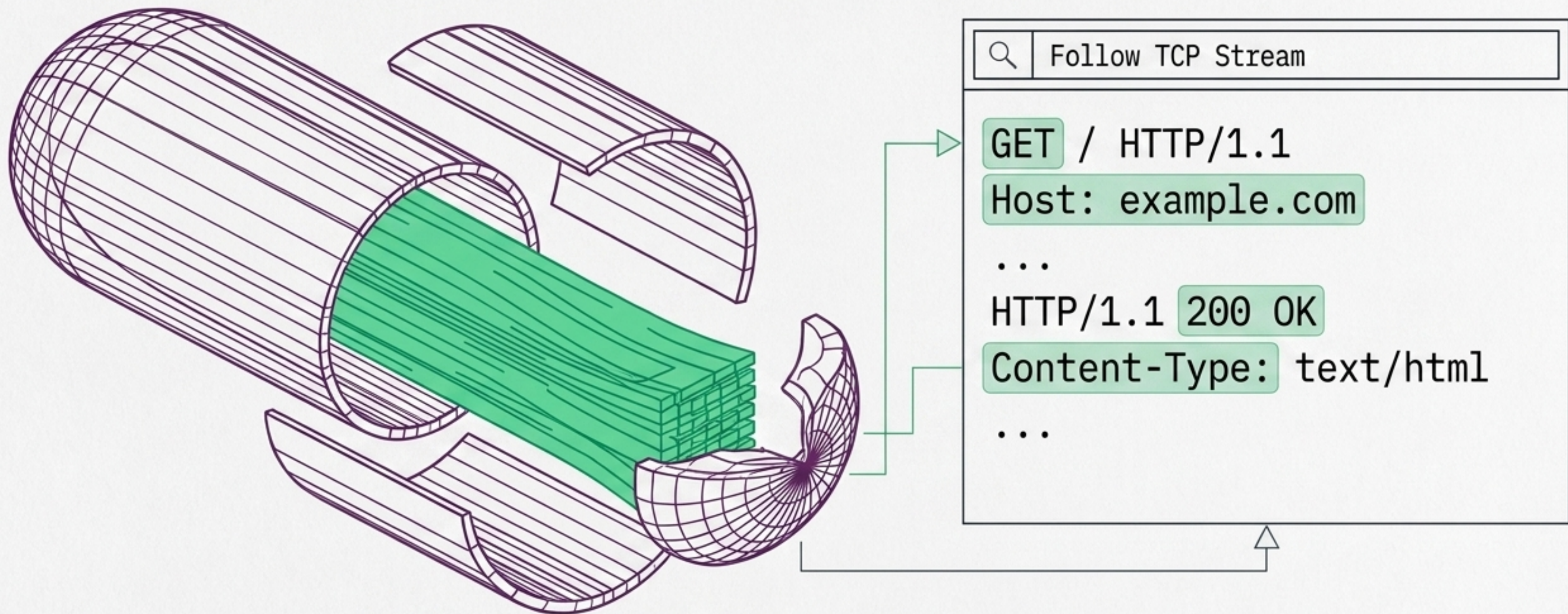


Securing the Payload in Transit



Delivering the Application Payload

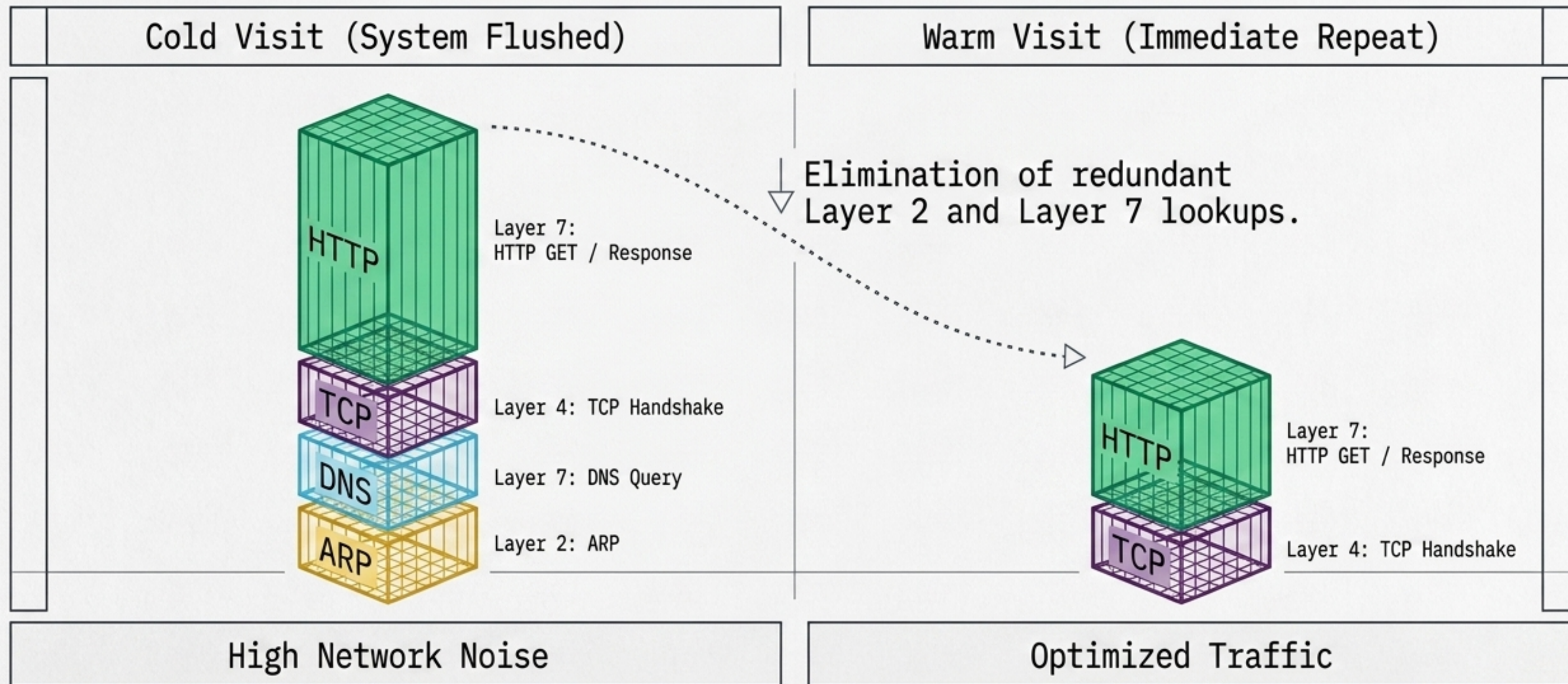
Reconstructing the Application Layer mechanics by extracting plain text from the transport wrapper.



Note: While modern web traffic is encrypted via HTTPS, legacy HTTP reveals the raw request/response cycle, serving as the ultimate architectural blueprint for application behavior.

The Invisible Efficiency of Caching

Empirical proof of how local caching fundamentally prevents internet congestion.



Diagnosing Infrastructure Failures

The Autopsy Part 1: Layers 2 and 3

	Failure Point	Symptom	Simulation Method	Diagnostic Tools	
	DNS Failure (L7 / L4)	Name resolution fails; Direct IP connections succeed.	resolv.conf pointed to black hole: 192.0.2.1	ping 192.0.2.1 nmap -p 53 192.0.2.1	
	Gateway / ARP Unreachable (L2 / L3)	Network completely unreachable; External pings fail.	Inject fake route with lower metric: ip route add default via 192.0.2.254 metric 50	ip route show ip neigh show arping	

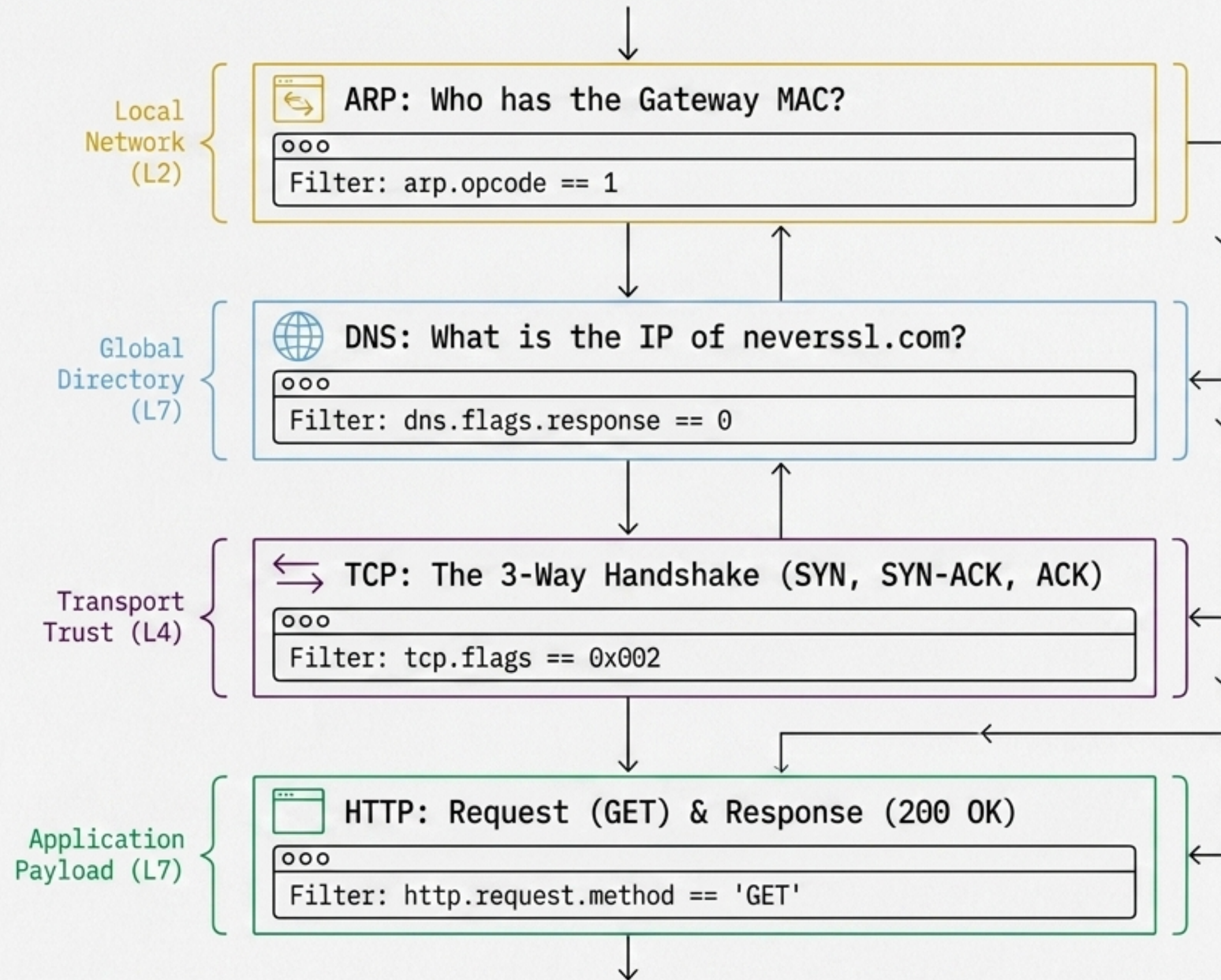
Diagnosing Transport and Application Failures

The Autopsy Part 2: Layers 4 and 7

Failure Point	Symptom	Diagnostic Signature / Insight
Firewall Drop (L4) 	Connection hangs, then times out.	Wireshark: SYN retransmissions with zero ACK responses. Simulation: iptables -A OUTPUT -p tcp --dport 80 -j DROP
TLS Handshake Failure (L7) 	Certificate Unknown / Mismatch.	Danger: Bypassing via curl -k in production disables cryptographic trust verification entirely.
HTTP 404 / 403 	Resource Not Found / Forbidden.	Network health is perfect. The TCP handshake succeeds, but the Application Server explicitly rejects the request.

The End-to-End Sequence of a Web Request

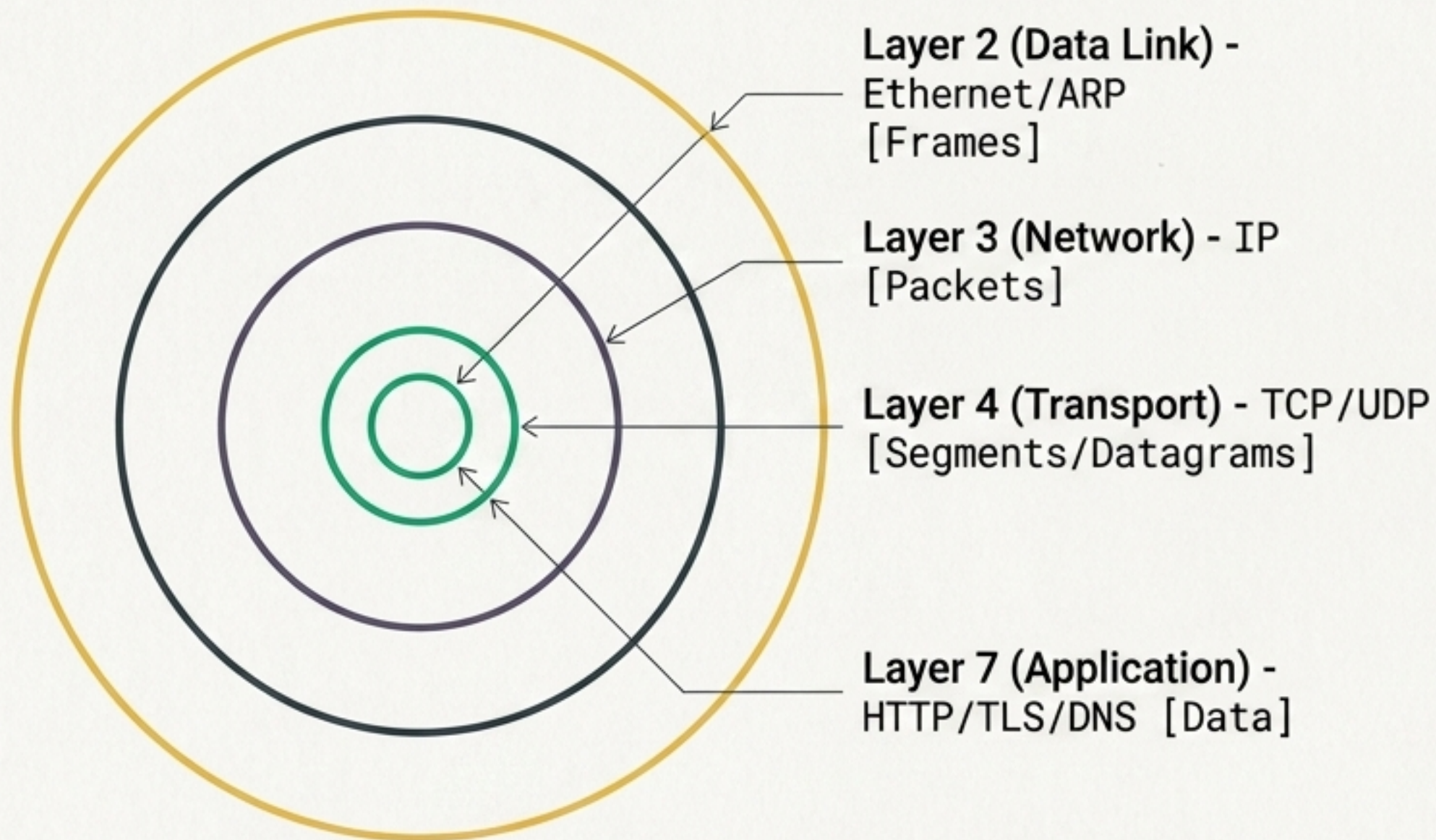
Capstone Diagram: A Complete Timeline of a Web Transaction



OSI Mapping and Attack Surfaces

Translating captured protocols into architectural layers and vulnerabilities

The Encapsulation Visual



The Security Matrix

Top Vector

ARP Poisoning (L2)

Exploits ARP's fatal flaw: blind trust without authentication.

Middle Vector

SYN Floods (L4)

Exploits the TCP handshake mechanism by abandoning half-open connections.

Bottom Vector

Session Hijacking (L7)

Steals application layer state after network trust is established.

The Digital Forensics Toolkit

Quick Reference Card

Network Configuration

<code>ip a</code>	→ Interfaces
<code>ip route show</code>	→ Routing Table
<code>ip neigh show</code>	→ ARP Cache

DNS & Connectivity

<code>dig <domain></code>	→ Detailed DNS lookup
<code>traceroute</code>	→ Hop-by-hop path
<code>curl -v</code>	→ Verbose HTTP inspection

Firewall Configuration

<code>sudo iptables -L -n</code>	→ List active rules
<code>iptables -A ... DROP</code>	→ Block specific port

Essential Wireshark Filters

<code>tcp.flags.syn == 1</code>	→ Isolate handshakes
<code>arp.opcode == 1</code>	→ ARP Requests
<code>http.request.method == 'GET'</code>	→ HTTP Requests
<code>tls</code>	→ Encrypted traffic